

ICP Final Project Report

Restaurant Ordering System

Name: Masoma Rafei

ID: 242261

AUW email: auw242261@auw.edu.bd

Name: Rayhan Mohammadi

ID: 242104

AUW email: auw242104@auw.edu.bd

Name: Malalay Rahmatzai

ID: 242264

AUW email: auw242264@auw.edu.bd

1. Introduction

The Restaurant Ordering System is a console-based C program designed to make the basic process of ordering food simple and organized. The idea behind the project is to simulate what happens when a customer enters a restaurant, looks through a menu, searches for a food item, places an order, and receives a final bill. I wanted the project to be practical rather than just a collection of small C programming exercises.

The program was developed with the main purpose of practicing the fundamental concepts of C programming in one complete project. It uses structures, arrays, functions, loops, conditions, pointers, searching, sorting, recursion, strings, and file handling. The interface is intentionally kept in the terminal so that the focus remains on the programming logic.

2. Main Features

The main menu is kept simple and contains only Customer Mode and Exit. Inside Customer Mode, the user can view the food menu, search for a particular food, sort the menu by price or name, and place an order. The program also checks available quantity in the background, so a customer cannot order more items than are available.

Feature	Purpose
View Menu	Displays food ID, name, category and price.
Search Food	Finds food using a name or part of a name.
Sort by Price	Arranges food from lower to higher price.
Sort by Name	Arranges food alphabetically.
Place Order	Allows multiple food items and quantities.
Discount	Automatically applies a discount based on subtotal.
Receipt	Shows itemized order and final payable amount.
File Handling	Keeps basic food data between program runs.

CS50.dev - Sample Program Output

```
=====
                        MAIN MENU
=====
1. Customer Mode
2. Exit
-----
Enter choice: 1

                        CUSTOMER MENU
1. View Food Menu
2. Search Food
3. Sort by Price
4. Sort by Name
5. Place Order
6. Back
Enter choice: 1

ID      Food                                     Category      Price
```

3. How the Program Works

When the program starts, it loads the available food information from a text file if that file exists. The user is then shown the main menu. Selecting Customer Mode opens a second menu. The customer can first inspect the menu or search for something specific. During ordering, the program asks for a table number, food ID, and quantity. Each selected item is stored in an order structure.

After the customer finishes ordering by entering 0, the program calculates the subtotal. A discount is then calculated according to the subtotal: 5% for orders of at least 500 BDT, 10% for orders of at least 1000 BDT, and 15% for orders of at least 1500 BDT. There is deliberately no tax in this version of the project. The final bill is simply the subtotal minus the discount.

4. C Programming Concepts Used

Structures: The program uses structures to represent food, cart items, and orders. This makes related information easier to manage.

Arrays: Food items and order items are stored in arrays, allowing the program to handle multiple records.

Functions: Separate functions are used for displaying the menu, searching, sorting, ordering, calculating bills, and saving data. This makes the code easier to read and maintain.

Pointers: Pointers are used when passing structures to functions, such as the order pointer used by the receipt and calculation functions.

Searching: The program searches food by ID and also searches by a word contained in the food name.

Sorting: Bubble sort is used to arrange food items by price and by name without changing the original menu order. **Recursion:** A small recursive function is used when the program starts, and another recursive function demonstrates recursive addition.

File handling: The program uses a text file named restaurant.txt to preserve food information such as price, stock, and sold quantity.

5. Sample Input and Output

For testing, I used a simple order containing two Chicken Biryani, one Beef Burger, and two Mango Juices. The subtotal is 770 BDT. Since the subtotal is above 500 BDT, the program gives a 5% discount of 38.50 BDT. The final amount is therefore 731.50 BDT. This test also shows that the program accepts multiple food items in a single order.

CS50.dev - Sample Order and Receipt

```
Enter choice: 5
Enter table number: 5
Enter Food ID (0 to finish): 1
Enter quantity: 2
2 x Chicken Biryani added.
Enter Food ID (0 to finish): 2
Enter quantity: 1
1 x Beef Burger added.
Enter Food ID (0 to finish): 9
Enter quantity: 2
2 x Mango Juice added.
Enter Food ID (0 to finish): 0
```

CUSTOMER RECEIPT

```
Order ID : 1001
Table    : 5
Chicken Biryani      2      360.00
Beef Burger          1      250.00
```

6. Conclusion

This project helped bring several C programming concepts together in one practical application. Instead of writing separate small programs for arrays, strings, searching, sorting, pointers, and recursion, the concepts are connected through a realistic restaurant scenario. The project can be expanded later with features such as customer accounts, different payment methods, table reservations, printable receipts, or a graphical interface. For the current project, however, keeping the program console-based makes the logic clear and suitable for a beginner-to-intermediate C programming assignment.